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By Amit Katiyar (MCA-JNU)



MATHEMATICS

- If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$ and $6A^{-1} = A^2 + CA + dI$, where A^{-1} is the inverse of A , I is the identity matrix, then (c, d) is
(a) (-6, 11) (b) (6, -11)
(c) (11, -6) (d) (6, 11)
- If $a = \hat{j} - \hat{k}$ and $c = \hat{i} - \hat{j} - \hat{k}$, then vector b satisfying $(a \times b) + c = 0$ and $a \cdot b = 3$ is
(a) $-\hat{i} + \hat{j} - 2\hat{k}$ (b) $2\hat{i} - \hat{j} + 2\hat{k}$
(c) $\hat{i} - \hat{j} - 2\hat{k}$ (d) $\hat{i} + \hat{j} - 2\hat{k}$
- Find the number of element in the union of four sets A, B, C and D having 150, 180, 210 and 240 elements respectively, given that each pair of sets has 15 elements in common. Each triple of sets has 3 elements in common and $A \cap B \cap C \cap D = \phi$
(a) 616 (b) 512
(c) 111 (d) 702
- If the straight line $ax + by + c = 0$ always passes through $(1, -2)$, then a, b and c are in
(a) AP (b) HP
(c) GP (d) None of these
- A six faced die is a biased one. It is thrice more likely to show an odd number than to show an even number. It is thrown twice. The probability that the sum of the numbers in the two throws is even, is
(a) $\frac{4}{8}$ (b) $\frac{5}{8}$ (c) $\frac{6}{8}$ (d) $\frac{7}{8}$
- If $I_n = \int_0^{\pi/4} \tan^n \theta d\theta$, then $I_8 + I_6$ is equal to
(a) $\frac{1}{4}$ (b) $\frac{1}{5}$ (c) $\frac{1}{6}$ (d) $\frac{1}{7}$
- If $\triangle ABC$ is a triangle whose area is $10\sqrt{3}$ units with side lengths $|AB| = 8$ units and $|AC| = 5$ units. Find the possible values of $\angle A$.
(a) 60° or 120° (b) 45° or 135°
(c) Only 30° (d) Only 90°
- Person A can hit a target 4 times in 5 attempts. Person B can hit a target 3 times 4 attempts person C can hit a target 2 times in 3 attempts. They fire a volley. The probability that the target is hit atleast two times, is
(a) $\frac{3}{4}$ (b) $\frac{1}{2}$ (c) $\frac{5}{6}$ (d) 1
- The value of the integral $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$ is
(a) 0 (b) $-\frac{\pi}{4}$ (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{4}$
- If ω is a cube root of unity, then find the value of
 $\begin{vmatrix} 1 + \omega & \omega^2 - \omega \\ 1 + \omega^2 & \omega - \omega^2 \\ \omega^2 + \omega & \omega - \omega^2 \end{vmatrix}$
(a) 3ω (b) -3ω (c) $3\omega^2$ (d) $-3\omega^2$
- If the vector $2\hat{i} - 3\hat{j}$, $\hat{i} + \hat{j} - \hat{k}$ and $3\hat{i} - \hat{k}$ form three conterminus edges of a parallelopiped, then the valume of parallelopiped is
(a) 8 (b) 10 (c) 4 (d) 14
- In a GP consisting of positive terms, each term equals the sum of the next two terms. Then, the common ratio of the GP is
(a) $\frac{(1-\sqrt{5})}{2}$ (b) $\frac{(\sqrt{5})}{2}$ (c) $\sqrt{5}$ (d) $\frac{(\sqrt{5}-1)}{2}$
- If $f(x) = \tan^{-1} \left[\frac{\sin x}{1 + \cos x} \right]$, then what is the first Derivative of $f(x)$?
(a) $\frac{1}{2}$ (b) $-\frac{1}{2}$ (c) 2 (d) -2

- The solution of $\sin x + 1 = \cos x$, such that $0 \leq x \leq 2\pi$, is
(a) 0, π (b) $0, \frac{\pi}{2}$ (c) $\frac{\pi}{2}, \frac{3\pi}{2}$ (d) $0, \frac{3\pi}{2}$
- If T_n denotes the number of triangle which can be formed by using the vertices of a regular polygon of n sides. If $T_{n+1} - T_n = 21$, then n equals
(a) 5 (b) 7 (c) 6 (d) 4
- If $\bar{x}_1$ and $\bar{x}_2$ the means of two distributions such that $\bar{x}_1 < \bar{x}_2$ and $\bar{x}$ is the mean of the combined distribution, then
(a) $\bar{x} < \bar{x}_1$ (b) $\bar{x} > \bar{x}_2$
(c) $\bar{x} = \frac{\bar{x}_1 + \bar{x}_2}{2}$ (d) $\bar{x}_1 < \bar{x} < \bar{x}_2$
- The area enclosed within the curve $|x| + |y| = 1$ (in sq units) is
(a) $\sqrt{2}$ (b) 1 (c) $\sqrt{3}$ (d) 2
- Let $f(x)$ be a polynomial function of second degree and $f(1) = f(-1)$. If a, b and c are in AP, then $f'(a), f'(b)$ and $f'(c)$ are in
(a) GP (b) HP (c) AGP (d) AP
- Find the point at which the tangent to the curve $y = \sqrt{4x - 3} - 1$ has its slope $\frac{2}{3}$.
(a) (3, 3) (b) (3, 2) (c) (2, 3) (d) (2, 2)
- Atal speaks truth in 70% and George speaks the truth in 60% cases. In what percentage of cases they are likely to contradict each other in stating the same fact?
(a) $\frac{13}{50}$ (b) $\frac{11}{50}$ (c) $\frac{23}{50}$ (d) $\frac{33}{50}$
- A man observes the angle of elevation of the top of a mountain to be 30° . He walks 1000ft nearer and finds the angle of elevation to be 45° . What is the distance of the first point of observation from the foot of the mountain?
(a) $500\sqrt{3}(\sqrt{3} + 1)$ ft (b) $500(\sqrt{3} + 1)$ ft
(c) $500(\sqrt{3} - 1)$ ft (d) $500\sqrt{3}(\sqrt{3} - 1)$ ft
- The sum of n terms is an arithmetic series is 216. The Value of the first term is n and the value of the n th term is $2n$. The common different d is
(a) 1 (b) $\frac{2}{3}$ (c) $\frac{3}{2}$ (d) $\frac{12}{11}$
- Force $3\hat{i} + 2\hat{j} + 5\hat{k}$ and $2\hat{i} + \hat{j} - 3\hat{k}$ are acting on a particle and displace it from the point $2\hat{i} - \hat{j} - 3\hat{k}$ to the point $4\hat{i} - 2\hat{j} + 7\hat{k}$, then the work done by the force is
(a) 18 units (b) 30 units (c) 27 units (d) 36 units
- The value of $9^{\frac{1}{3}}, 9^{\frac{1}{9}}, 9^{\frac{1}{27}} \dots$ is
(a) 3 (b) 6 (c) 9 (d) None
- The minimum value of the function $y = 2x^3 - 21x^2 + 36x - 20$ is
(a) -120 (b) -126 (c) -128 (d) None of these
- In how many different ways can the letters of the word 'CORPORATION' be arranged, so that all the vowels always come together?
(a) 810 (b) 1440 (c) 2880 (d) 50400
- If $\log_x y = 100$ and $\log_2 x = 10$, then the value of y is
(a) 2^{10} (b) 2^{100} (c) 2^{1000} (d) 2^{10000}
- The equation of the line parallel to the line $2x - 3y = 7$ and passing through the mid-point of the line segment joining the point (1, 3) and (1, -7) is
(a) $2x - 3y - 4 = 0$ (b) $2x - 3y + 4 = 0$ (c) $2x - 3y - 8 = 0$ (d) $2x - 3y + 8 = 0$



29. In a $\triangle ABC$, if $(c + a + b)(a + b - c) = ab$, then measure of $\angle C$ is
 (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$
 (c) $\frac{2\pi}{3}$ (d) None of these
30. The number of none-negative integers less than 1000 that contain the digit 1 are
 (a) 9^2 (b) 9^3
 (c) $10^2 - 9^2$ (d) $10^3 - 9^3$
31. The lines $3x - 4y + 4 = 0$ and $6x - 8y - 7 = 0$ are tangent to the same circle. The radius of this circle is
 (a) $\frac{3}{2}$ (b) $\frac{3}{4}$
 (c) $\frac{4}{5}$ (d) $\frac{7}{10}$
32. The area of the parallelogram whose diagonals are $a = 3\hat{i} + \hat{j} - 2\hat{k}$ and $b = \hat{i} - 3\hat{j} + 4\hat{k}$ is
 (a) $10\sqrt{3}$ (b) $5\sqrt{3}$
 (c) $10\sqrt{2}$ (d) $5\sqrt{2}$
33. If $\sin x + a \cos x = b$, then what is the expression for $|a \sin x - \cos x|$ in terms of a and b ?
 (a) $a = \sqrt{a^2 - b^2 - 1}$ (b) $\sqrt{a^2 + b^2 - 1}$
 (c) $\sqrt{a^2 + b^2 + 1}$ (d) $\sqrt{a^2 - b^2 + 1}$
34. If A and B are two events such that $P(A \cup B) = \frac{5}{6}$, $P(A \cap B) = \frac{1}{2}$ and $P(\bar{B}) = \frac{1}{2}$, then the events A and B are
 (a) Dependent (b) Independent
 (c) Mutually exclusive (d) None of these
35. If the three vectors $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} + 2\hat{j} - 3\hat{k}$ and $3\hat{i} + \lambda\hat{j} + 5\hat{k}$ are coplanar, then λ is
 (a) -1 (b) -2
 (c) -3 (d) -4
36. The equation of the base of an equilateral triangle is $x + y = 2$ and the vertex is $(2, -1)$. The length of the side of the triangle is
 (a) $\sqrt{\frac{3}{2}}$ (b) $\sqrt{2}$
 (c) $\sqrt{\frac{2}{3}}$ (d) $\sqrt{\frac{20}{3}}$
37. The total number of numbers that can be formed using the digit 3, 5 and 7 only, if no repetitions are allowed, is
 (a) 39 (b) 105
 (c) 15 (d) 27
38. If $x = a \cos t$, $y = b \sin t$, then $\frac{d^2y}{dx^2}$ is
 (a) $-\frac{b^4}{a^2y^3}$ (b) $\frac{b^4}{a^2x^3}$
 (c) $\frac{b}{ay^4}$ (d) $-\frac{a^4}{bx^3}$
39. A random variable x has the distribution law as given below:
- | | | | |
|------------|-----|-----|-----|
| x | 1 | 2 | 3 |
| $P(X = x)$ | 0.3 | 0.4 | 0.3 |
- The variance of the distribution is
 (a) 0.4 (b) 0.6
 (c) 2 (d) None of these
40. The value of $\tan \theta + 2 \tan 2\theta + 4 \tan 4\theta + 8 \cot 8\theta$ is
 (a) $\cot \theta$ (b) $\tan \theta$
 (c) $\sin \theta$ (d) $\cos \theta$
41. The sum of integers between 200 and 400, that are multiples of 7 is
 (a) 8729 (b) 8700
 (c) 8972 (d) 8279

42. $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^2 \tan x}$ is equal to
 (a) 0 (b) 1 (c) $\frac{1}{2}$ (d) $\frac{1}{3}$
43. Two fair dice are tossed. What is the probability that the total score is a prime number?
 (a) $\frac{1}{6}$ (b) $\frac{5}{12}$ (c) $\frac{1}{2}$ (d) $\frac{7}{9}$
44. Find the equation of the circle which passes through $(-1, 1)$ and $(2, 1)$ and having centre on the line $x + 2y + 3 = 0$.
 (a) $2x^2 + 2y^2 - 2x + 7y - 13 = 0$
 (b) $x^2 + y^2 - 2x + 7 - 13 = 0$
 (c) $2x^2 + 2y^2 + 2x + 7y - 13 = 0$
 (d) $x^2y^2 + 2x + 7y - 13 = 0$
45. If a, b and c are the position vector of three vertices A, B and C of a triangle, respectively. Then, the area of this triangle is given by
 (a) $\frac{1}{2}(a \times b) \cdot c$ (b) $\frac{1}{2}|a \times b + b \times c + c \times a|$
 (c) $a \times b + b \times c + c \times a$ (d) None of these
46. The sum of the focal distances of any point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with eccentricity e , is given by
 (a) $2ae$ (b) $2b$ (c) $2a$ (d) $2be$
47. If $\sin x + \sin^2 x = 1$, then $\cos^2 x + \cos^4 x$ is equal to
 (a) 0 (b) 1 (c) -1 (d) 2
48. An experiment succeeds twice often as it fails. The probability that in the next six trials there will be atleast four successes, is
 (a) $\frac{240}{729}$ (b) $\frac{496}{729}$
 (c) $\frac{220}{729}$ (d) $\frac{233}{729}$
49. Sum of 20 terms of the series $-1^2 + 2^2 - 3^2 + 4^2 - \dots$ is
 (a) 180 (b) 200 (c) 210 (d) 220
50. If $\tan \alpha = \frac{m}{m+1}$ and $\beta = \frac{1}{2m+1}$, then $\alpha + \beta$ is equal to
 (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{6}$ (d) π

REASONING

51. A train takes 18 s to pass completely through a station 162 m long and 15s through another station 120 m long, at the same speed. What is then length of the train in metre?
 (a) 70 (b) 80 (c) 90 (d) 100
52. In a row of children facing North, Shamika is third to the right of Nikhil, who is 17th from the right end of the row. Ravi is 5th to the left of Shamika and is 20th from the left end. Totally how many children are there in the row?
 (a) 37 (b) 38
 (c) 39 (d) None
53. Given that
- Some apples are blackberries.
 - Some doughnuts are apples.
 - No coconut is a doughnut.
 - All blackberries are coconuts.
- Which of the following statements is false?
 (a) Some blackberries are doughnut
 (b) Some coconut are apples
 (c) All coconuts are not apples
 (d) All doughnuts are not coconuts



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Direction[Q. Nos.54-56]In a family of 6 persons, there are two couples. The lawyer is the head of the family and has two sons –Mukesh and rakesh both teachers.Mrs. Reena and her mother-in-law both are lawyers. Mukesh's wife is a doctor and they have a son, Ajay.

54. Which of the following is definitely a couple?

- (a)Lawyer-Teacher
- (b)Doctor-Lawyer
- (c)Teacher-Teacher
- (d)None of these

55. What is the profession of Rakesh's wife?

- (a)Teacher
- (b)Lawyer
- (c)Doctor
- (d)Cannot be determind

56. What is/was Ajay's grandfather's occupation ?

- (a)Teacher
- (b)Lawyer
- (c)Doctor
- (d)Cannot be dertermind

57. Find the missing element is the series.

A, CD, GHI, ..., UVWXY

- (a)LMNO
- (b)MNOP
- (c)NOPQ
- (d)QPQR

58. In a code language, 'FRIEND' is coded as 'GTLIST'. Which of the following is coded as 'HWDVT' in that language?

- (a)HAPPY
- (b)GUARD
- (c)BEADS
- (d)SPEED

59. There are four brother Alan, Bob, Carl and Dave. Dave is two years older than Bob. Bob is one year younger than Carl. Alan, who is 34, is two years younger than Carl. Who is the oldest ?

- (a)Alan
- (b)Bob
- (c)Carl
- (d)Dave

Directions-Q. Nos. 60-62: An employee has been assigned the task of allotting offices to six of the staff members. The offices are numbered 1-6. The offices are arranged in a row and they are separated from each other by six foot high dividers. Hence, voices sounds and cigarette smoke flow easily from one office to another.

Miss. Robert needs to use the telephone quite often throughout the day. Mr. Mike and Mr. Brown need adjacent offices as they need to consult each other often while working. Miss. Hardy is a senior employee and has to be allotted the office number 5, having the biggest window. Mr. Donald requires silence in the offices next to his and Mr. Tim prefers to be as away as possible from Miss. Robert. Mr. Donald are all smokers. Miss. Hardy finds tobacco smoke allergic and consecutively the offices next to hers to be occupied by non-smokers. Unless specifically stated all the employees maintain an atmosphere of silence during office hours.

60. The ideal candidate to occupy the office farthest from Mr. Brown would be

- (a)Miss. Hardy
- (b)Mr. Mike
- (c)Mr. Tim
- (d)Mr. Donald

61. The three employees who are smokers, should be seated in the offices

- (a)1, 2 and 4
- (b)2, 3 and 6
- (c)1, 2 and 6
- (d)1, 2and 3

62. The ideal office for Mr. Mike would be

- (a)2
- (b)6
- (c)1
- (d)3

63. A doctor said to his compunder 'I go to see the patients at their residence after every 3 h 30 min. I have already gone to the patient 1 h 20 min ago and next time I shall go at 1:40 pm. 'At what time this information was given to the compunder by the doctor?

- (a)11:30am
- (b)11:20 am
- (c)10:10 am
- (d)None of these

64. Which pair of number comes next in the following series?

42 40 38 35 33 31 28

- (a)25 22
- (b)26 23
- (c)26 24
- (d)25 23

Directions [Q. Nos. 65-66]

- i. All G's are H's
- ii. All G's are J's or K's
- iii. All J's are K's are G's
- iv. All L's are K's
- v. All N's are M's
- vi. No M's are G's

65. If no P's are K's ,then which of the following must be true?

- (a)All P's are J's
- (b)If any P is a G, it is a J
- (c) No P is an H
- (d) If any P is an H,it is a G

66. Which of the following is inconsistent with one or more of the conditions?

- (a)All H's are G's
- (b)All H's that are not G's are M's
- (c)Some H's are both M's and G's
- (d)No M's are H's

67. Shyam is taller than Pradeep and Pradeep is as tall as Anurag. But Anand is shorter than Suresh, who is as tall as Anurag. If If Pradeep is taller than Praveen,then who is the tallest of all?

- (a)Pradeep
- (b)Praveen
- (c)Suresh
- (d)Shyam

68. When Rajveer was born his father was 32yr older than his brother and his mother was 25yr older than his sister. If Rajeev's brother is 6 yr older than Rajeev and his mother is 3yr younger than his father, then how old was Rajeev's sister when he was born?

- (a) 15yr
- (b) 14yr
- (c) 7 yr
- (d) 10yr

69. Dhoni starts from his office at 8 am on a Sunday morning, travels 10 km towards 10km towards West and then turns to his left and walks 8 km. Then ,he again turns to his left and walks 4 km and then stops. What is the shortest distance to his office from the point ,where he stopped?

- (a) 18 km
- (b)8km
- (c) 10 km
- (d)None

70. A treasure chest has less than 100 gold coins. The number of coins is

- (a) One more than a multiple of 3
- (b) Two more than a multiple of 4
- (c) Three more than a multiple of 5 and
- (d) Four more than a multiple of 6

How many coins are there in the chest?

- (a)58
- (b) 88
- (c) 98
- (d) 38

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71. Read the statements and then decide which of the conclusions logically follow:

Statements

- (i) All mangoes are golden in colour.
- (ii) No golden coloured things are cheap.

Conclusions

- (i) All mangoes are cheap.
- (ii) Golden coloured mangoes are not cheap.
- a) Only conclusion I follows
- b) Only conclusion II follows
- c) Either I or II follows
- d) Neither I nor II follows

Directions [Q. Nos. 72-73]: A blacksmith has five iron articles A,B,C,D and E, each having a different weight.

- (a) Weighs twice as much as B.
- (b) Weighs four and half times as much as C.
- (c) Weighs half as much as E.
- (d) Weighs less than A but more than C.

72. Which of the following article is heaviest in weight

- (a) A
- (b) B
- (c) C
- (d) D

73. Which of the following represents the descending order of weights of the articles?

- (a) A,B,E,D,C
- (b) A,B,C,D,E
- (c) B,D,E,A,C
- (d) C,D,E,B,A

Directions [Q.nos. 74-76]: There are three switches A,B and C which can be in ON/OFF position. Their settings change as per the following rules:

- (i) If A is the only switch as ON, change B to ON.
- (ii) If A and B are only switches as ON, change C to ON.
- (iii) If all three switches are ON, change C to OFF.
- (iv) For all other situations, all switches in On are changed to OFF and all switches in OFF are changed to ON.

74. If switches A and B are ON and C is OFF, then their changed settings will be

- (a) A ON, B OFF, C OFF
- (b) A ON, B ON, C ON
- (c) A ON, B OFF, C ON
- (d) A OFF, B ON, C OFF

75. If only B is ON, then the changed setting, then which of the following could have been the original setting?

- (a) A ON, B ON, C ON
- (b) A ON, B ON, C OFF
- (c) A ON, B OFF, C ON
- (d) A OFF, B OFF, C ON

76. If only B is ON in the changed setting, then which of the following could have been the original setting?

- (a) A ON, B OFF, C OFF
- (b) A ON, B ON, C ON
- (c) A ON, B OFF, C ON
- (d) A OFF, B OFF, C OFF

77. If the third day of a month falls on Friday, then what day will be on the fourth day after twenty first of the month?

- (a) Monday
- (b) Tuesday
- (c) Saturday
- (d) Thursday

78. Ana is a girl and has the same number of brothers and sisters. Andrew is a boy and has twice as many sisters as brothers. Ana and Andrew are the children of Emma.

How many children does Emma have?

- (a) 2
- (b) 3
- (c) 5
- (d) 7

Directions [Q. Nos. 79-81]

1. Anu is taller than Cini.
 2. Eenu is shorter than Binu.
 3. Anu is shorter than Danny.
 4. Eenu is taller than Anu.
79. The best answer to 'who is the tallest?' is
- (a) Dany
 - (b) Binu
 - (c) Dany or Binu
 - (d) Both Dany and Binu
80. Who is the shortest?
- (a) Cini
 - (b) Anu or Cini
 - (c) Eenu
 - (d) Insufficient data to conclude
81. Which of the following statements would help to logically order the persons according to their heights?
- (a) Binu is 7 feet tall
 - (b) Dany and Binu do not have equal height
 - (c) Eenu is the tallest in the group
 - (d) Dany is the tallest in the group
82. Karan and Arjun run a 100 m race, where Karan beats Arjun by 10 m. To do a favour of Arjun, Karan starts 10 m behind the starting line in a second 100 m race. They both run at their earlier speeds. Which of the following is true in connection with the second race?
- (a) Karan and Arjun reach the finishing line simultaneously
 - (b) Karan beats Arjun by 1 m
 - (c) Arjun beats Karan by 11 m
 - (d) Arjun beats Karan by 1 m
83. In a cricket season, India defeated Australia twice, West Indies defeated India twice. Australia defeated West Indies twice. India defeated New Zealand twice and West Indies defeated New Zealand twice. Which country has lost most number of times?
- (a) India
 - (b) Australia
 - (c) New Zealand
 - (d) West Indies
84. Pointing to a woman, Nirmal said, "She is the daughter of my wife's grandfather's only child". How is the woman related to Nirmal?
- (a) Wife
 - (b) Sister-in-law
 - (c) Sister
 - (d) None of these
- Direction** [Q. Nos. 85-87] These are five persons A, B, C, D and E standing on six steps numbered 1, 2, 3, 4, 5 and 6 from the bottom. At most one person is standing on each step. The step number on which A is standing, is two less than that of C. Step number on which B is standing is one more than that of D.
85. If A is standing on step 1, which of the following is true?
- (a) B is standing on step 2
 - (b) C is standing on step 4
 - (c) E is standing on step 3
 - (d) D is standing one step higher than C
86. If D is standing on step 1, on which step A could be standing?
- (a) 2 or 4 only
 - (b) 3 or 5 only
 - (c) 3 or 4 only
 - (d) 4 or 5 only
87. If there are two steps in between the steps on which A and D are standing, A must be standing on which of the following steps?
- (a) 3
 - (b) 4
 - (c) 5
 - (d) 6

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88. From the information given below:

A*B means A and B are of the same age.

A—B means B is younger than A.

A+B means A is younger than B.

What does Sachin * Mohan—Ravi mean?

(a)Sachin is youngest

(b)Ravi is youngest

(c)Sachin is oldest

(d)Mohan is oldest

89. Jimmy saw the time while going to the tennis court. He saw the hour hand is 20° away from 4. After he returned from tennis court, he noticed that the hour hand is 20° away from 4. If he took ten minutes to go to tennis court and he walked at the same speed while going to the tennis court and while returning, how much time did he spent at the tennis court?

(a)60 min

(b)80 min

(c)70 min

(d)50 min

90. There are 8 balls looking alike, seven of which have equal weight and one is slightly heavier. The weighing balance is of unlimited capacity. Using this balance, the minimum number of weighings required to identify the heavier ball is

(a)1

(b)2

(c)3

(d)4

GENERAL ENGLISH

91. Out of the alternatives, choose the appropriate phrase to make the sentence meaningful. If you had joint accounts with who died, then you will be responsible for the bills.

(a)everybody

(b)anyone

(c)everyone

(d)someone

92. Choose the analogy that is closest in meaning to the pair:

Diamond : Necklace

(a)Car : Roads

(b)Flowers : Bouquet

(c)Gold : Bangle

(d)Book : Shop

93. Choose the suitable preposition for the blank to make a meaningful sentence. Suresh is angryhis servant.

(a)about

(b)on

(c)by

(d)with

94. Choose the correct alternative for the sentence below: The Earth is always revolving round the Sun.

(a)The Earth revolves round the sun

(b)The Earth is revolving round the Sun

(c)The Earth revolving round the Sun

(d)None of the above

95. Choose the word that best expresses the meaning of the given idiom: 'A close shave.'

(a)A clean shave

(b)A narrow escape

(c)A guarded secret

(d)A sudden fall

96. Pick the part of the sentence that has an error.

My elder brother is a MA whereas I am only a BA

(a)My elder brother

(b) is a MA

(c)whereas I am

(d)only a BA

97. Choose the suitable phrasal verb for the blank in the sentence below:

I _____my hopes when untimely rain threatened my crops.

(a)gave in

(b)gave out

(c)gave up

(d)gave off

98. Out of the given alternatives, choose the word that is opposite in meaning to the word

AFFLUENT

(a)Reluctant

(b)Poor

(c)Clear

(d)Enthusiastic

99. Fill in the blank with appropriate form of noun:

Don't blame yourself, it's not your.....

(a)misunderstanding

(b)error

(c)slip

(d)fault

100. Fill in the blanks:

The instruction, alongwiththe class angry about the room change.

(a)are

(b)have

(c)has

(d)is

101. Choose the suitable word for the blank to make it a meaningful statement.

What you say is..... my comprehension?

(a)before

(b)beside

(c)behind

(d)beyond

102. Fill in the blank with a suitable preposition:

If you want to avoid traffic, you need to leave _____

(a)until

(b)by

(c)during

(d)at

103. Choose the word the best expression the meaning of the given idiom: 'To smell a rat'.

(a)To suspect something bad

(b)To misunderstanding

(c)To detect bad smell

(d)To forsake

104. Out of the given alternatives, choose the word that best expresses the meaning of the word abridge.

(a)Judge

(b)Release

(c)Shorten

(d)Dissolve

105. 'A dog's breakfast means

(a)breakfast cooked for a dog

(b)breakfast cooked by a dog

(c)something that has been done very badly

(d)None of the above

106. Change the speech:

She says, "I like going to the sea side".

(a)She says she likes going to the sea side

(b)She says I like going to the sea side

(c)She says that she liked going to the sea side

(d)She says she like going to the sea side

107. Arrange the following to form a correct sentence.

P : will normally be granted

Q : candidates should note

R : that no request for

S : change of centre

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- (a)SRQP (b)PRQS
(c)QSPR (d)QRSP

108. Rewrite the sentence after correcting the error.

She was one of the average student of the class

- (a)She was one of the average students of the class
(b)She is one of the average students of the class
(c)She was one among the average student of the class
(d)She is an average students of the class

109. Choose appropriate words to form a grammatically correct sentence.

The decoration of the new house, including the furniture and curtains.....

- (a)is more pleasing (b)are more pleasing
(c)is most pleasing (d)are pleasing

110. Fill in the blank:

The President of the United States, accompanied by his advisor.....enroute to Europe.

- (a)were (b)are
(c)was (d)Both (a) and (c)

COMPUTER

111. All digit circuits can be realised by using only

- (a)Exclusive OR gates (b)Half adder
(c)Multiplexers (d)OR gate

112. The Boolean function $a + (a \cdot b)$ is equivalent to

- (a) $a \cdot b$ (b) $a + b$
(c) $a \cdot \bar{b}$ (d) $\bar{a} + b$

113. Which of the following circuit is used as a memory device in computers?

- (a)Flip-Flop (b)Rectifier
(c)Comparator (d)All of these

114. Convert the hexadecimal number 4DF to its octal equivalent.

- (a)2333 (b)2337
(c)2773 (d)2373

115. A tautology is a Boolean formula that is always true.

Which of the following is a tautology?

- (a)x (b) $(\bar{x} + \bar{x})y$
(c) $x + \bar{y} + \bar{x}$ (d) $(xy) + \bar{x}$

116. Acronym of EEPROM is

- (a)Extended Erasable Programmable Memory
(b)Electrically Erasable Read Only Memory
(c)Electrically Erasable Programmable Read Only Memory
(d)Extended Erasable Page-oriented Memory

117. For reproducing sound, a CD audio player uses a

- (a)Quartz crystal (b)Titanium
(c)Barium ceramic (d)Laser beam

118. When we open an internet site, we see WWW. What does WWW Stand for ?

- (a)World Wide Word (b)World Wide Web
(c)World Wide Webinar (d)Word Widing Works

119. The answer of the operation $(10111)_2 \times (1110)_2$ in hexaequivalent is

- (a)150 (b)14 C
(c)142 (d)13 E

120. The minimum number of bits to represent character from ASCII code set is

- (a)2 (b)8
(c)5 (d)7





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MAARULA MATHEM –E LEARNING

ANSWER KEY

1. (a)	21.(a)	41.(a)	61.(d)	81.(d)	101. (d)
2. (a)	22.(d)	42.(d)	62.(d)	82.(d)	102. (b)
3. (d)	23.(c)	43.(b)	63.(a)	83.(c)	103. (a)
4. (a)	24.(a)	44.(a)	64.(c)	84.(a)	104. (c)
5. (b)	25.(d)	45.(b)	65.(b)	85.(d)	105. (c)
6. (d)	26.(d)	46.(c)	66.(c)	86.(c)	106. (a)
7. (a)	27.(c)	47.(b)	67.(d)	87.(b)	107. (d)
8. (c)	28.(c)	48.(b)	68.(d)	88.(b)	108. (a)
9. (d)	29.(c)	49.(c)	69.(c)	89.(a)	109. (c)
10. (d)	30.(d)	50.(b)	70.(a)	90.(c)	110. (c)
11. (c)	31.(b)	51.(c)	71.(b)	91.(b)	111. (b)
12. (d)	32.(a)	52.(b)	72.(a)	92.(c)	112. (b)
13. (a)	33.(d)	53.(a)	73.(a)	93.(d)	113. (a)
14. (d)	34.(b)	54.(a)	74.(b)	94.(a)	114. (b)
15. (b)	35.(d)	55.(c)	75.(c)	95.(b)	115. (c)
16. (d)	36.(c)	56.(d)	76.(b)	96.(b)	116. (c)
17. (d)	37.(c)	57.(b)	77.(c)	97.(c)	117. (d)
18. (d)	38.(a)	58.(b)	78.(d)	98.(b)	118. (b)
19. (b)	39.(b)	59.(d)	79.(c)	99.(d)	119. (c)
20. (c)	40.(a)	60.(d)	80.(a)	100. (d)	120. (d)

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SOLUTION

Maths

1. $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$

and $6A^{-1} = A^2 + CA + dI$ (1)

Now multiplying Eq. (i) by A,

$6I = A^3 + CA^2 + dA$

$A^3 + CA^2 + dA - 6I = 0$(2)

Applying Cayley Hamilton rule

$$\begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 1-\lambda & 1 \\ 0 & -2 & 4-\lambda \end{vmatrix}$$

$\Rightarrow (1-\lambda) \{(1-\lambda)(4-\lambda) 2\} = 0$

$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$(3)

on comparing Eqs (ii) and (iii)

$c = -6$ and $d = 11$

2. (a) $(a \times b) + c = 0$

$a \times (a \times b) + a \times c = 0$

$(a \cdot b)a - (a \cdot a)b + a \times c = 0$ (i)

$$a \times c = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & -1 \\ 1 & -1 & -1 \end{vmatrix} = 2\hat{i} - \hat{j} - \hat{k}$$

As, $a = \hat{j} - \hat{k}$, $c = \hat{i} - \hat{j} - \hat{k}$

Put these values in Eq. (i)

$3(\hat{j} - \hat{k}) - 2\hat{i} - \hat{j} - \hat{k} = (a \cdot a)b$

$2b = -2\hat{i} + 2\hat{j} - 4\hat{k}$

$b = -\hat{i} + \hat{j} - 2\hat{k}$

3. (d) $n(A) = 150, n(B) = 180, n(C) = 210, n(D) = 240$

$n(A \cap B) = n(B \cap C) = n(C \cap D) = n(A \cap D)$

$= n(A \cap C) = n(B \cap D) = 15$

$n(A \cap B \cap C) = n(B \cap C \cap D) = n(A \cap C \cap D)$

$= n(A \cap B \cap D) = 3$

$n(A \cap B \cap C \cap D) = \phi$

$n(A \cup B \cup C \cup D) = n(A) + n(B) + n(C)$

$- n(A \cap B \cap C \cap D)$

$= 150 + 180 + 210 + 240 - 6 \times 15 + 4 \times 3 - 0 = 702$

4. $ax + by + c = 0$ passes through (1, -2)

$a - 2b + c = 0$

$2b = a + c$

a, b, c are in AP

5. (b)

Odd number be event A

Even no event B

$P(A) = 3 P(B)$

Let $P(A) = P$

$P = 3p$

Pt $3p = 1$

$4P = 1, P = \frac{1}{4}$

Possible chances

Even in 10^1 chance and a x a in 2^{nd} chances

$= \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$

Odd in 1^{st} throw and odd in 2^{nd} chance

$\frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$

Total probability $= \frac{9}{16} + \frac{1}{16}$

$= \frac{10}{16} = \frac{5}{8}$

6. (d) $I_8 = \int_0^{\pi/4} \tan^8 \theta d\theta, I_6 = \int_0^{\pi/4} \tan^6 \theta d\theta$

$I_8 + I_6 = \int_0^{\pi/4} \tan^8 \theta d\theta + \int_0^{\pi/4} \tan^6 \theta d\theta$

$= \int_0^{\pi/4} \tan^6 \theta (1 + \tan^2 \theta) d\theta = \int_0^{\pi/4} \tan^6 \theta \sec^2 \theta d\theta$

Let $\tan \theta = t \Rightarrow \sec^2 \theta d\theta = dt = \int_0^1 t^6 dt$

$I_8 + I_6 = \left[\frac{t^7}{7} \right]_0^1 = \frac{1}{7} - 0 = \frac{1}{7}$

7. (a) Area of $\triangle ABC = 10\sqrt{3}$

We know for $\triangle ABC$

Area $= \frac{1}{2} ac \sin B = \frac{1}{2} bc \sin A = \frac{1}{2} casin B$

Here, $c = 8, b = 5 \Rightarrow 10\sqrt{3} = \frac{1}{2} \times 8 \times 5 \times \sin A$

$\frac{4\sqrt{3}}{8} = \sin A \Rightarrow \sin A = \frac{\sqrt{3}}{2}, A = \frac{\pi}{3}, \frac{2\pi}{3} \Rightarrow A = 60^\circ, 120^\circ$

8. (c) Probability that A hit the target is $4/5$.

Probability that B hit the target is $3/4$.

Probability that C hit the target is $2/3$.

Probability that target is hit at least 2 times.

$[ABC + \bar{A}BC + A\bar{B}C + ABC \text{ (atleast two times)}]$

$\frac{4}{5} \times \frac{3}{4} \times \frac{1}{3} + \frac{4}{5} \times \frac{1}{4} \times \frac{2}{3} + \frac{1}{5} \times \frac{3}{4} \times \frac{2}{3} + \frac{4}{5} \times \frac{3}{4} \times \frac{2}{3}$

$= \frac{1}{5} + \frac{2}{15} + \frac{1}{10} + \frac{2}{5}$

$= \frac{50}{60} = \frac{5}{6}$

9. (d) $I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$

Using shortcut formula $\int_a^b \frac{f(x)}{f(x) + f(a+b-x)} dx = \frac{b-a}{2}$

$\Rightarrow I = \frac{\pi-0}{2} \Rightarrow I = \frac{\pi}{4}$

10. (d). Since, ω is cube root of unity. $1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$

Now, $\begin{vmatrix} 1 + \omega & \omega^2 & -\omega \\ 1 + \omega^2 & \omega & -\omega^2 \\ \omega^2 + \omega & \omega & -\omega^2 \end{vmatrix}$

$C_1 \rightarrow C_1 + C_2$

$= \begin{vmatrix} 0 & \omega^2 & -\omega \\ 0 & \omega & -\omega^2 \\ \omega^2 + 2\omega & \omega & -\omega^2 \end{vmatrix}$

$R_3 \rightarrow R_3 - R_2$



$$= \begin{vmatrix} 0 & \omega^2 & -\omega \\ 0 & \omega & -\omega^2 \\ \omega^2 + 2\omega & 0 & 0 \end{vmatrix}$$

$$= (\omega^2 + 2\omega)(-\omega^4 + \omega^2) = (\omega^2 + \omega + \omega)(-\omega + \omega^2)$$

$$= (-1 + \omega)(-\omega + \omega^2) = \omega - \omega^2 - \omega^2 + \omega^3$$

$$= 1 + \omega - 2\omega^2 = -\omega^2 - 2\omega^2 = -3\omega^2$$

11. (c) Volume of parallelepiped = $[a \ b \ c]$

Volume =

$$= 2(-1) + 3(-1 + 3) + 0 = -2 + 3(2) = -2 + 6 = 4$$

Volume = 4

12. (d) Let the term of GP be $a, ar, ar^2, \dots$

According to question,

$$a = ar + ar^2 \Rightarrow a = a(r + r^2)$$

$$r^2 + r - 1 = 0$$

$$r = \frac{-1 \pm \sqrt{1+4}}{2} \Rightarrow r = \frac{-1 \pm \sqrt{5}}{2}$$

Since, 'r' is positive.

Hence,

$$r = \frac{-1 + \sqrt{5}}{2}$$

$$13. y = \tan^{-1} \left(\frac{\sin x}{1 + \cos x} \right)$$

$$y = \tan^{-1} \left(\frac{\sin \left(\frac{2x}{2} \right)}{2 \cos^2 \left(\frac{x}{2} \right)} \right)$$

$$y = \tan^{-1} \left(\frac{2 \sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right)}{2 \cos^2 \left(\frac{x}{2} \right)} \right)$$

$$y = \tan^{-1} \left(\frac{\sin \left(\frac{x}{2} \right)}{\cos \left(\frac{x}{2} \right)} \right)$$

$$y = \tan^{-1} \left(\tan \left(\frac{x}{2} \right) \right)$$

$$y = x/2$$

$$\frac{dy}{dx} = \frac{1}{2}$$

14. (d)

$$\sin x + 1 = \cos x, 0 \leq x \leq 2\pi$$

$$\Rightarrow \sin x - \cos x = -1$$

$$\Rightarrow \sqrt{2} \left(\frac{1}{\sqrt{2}} \sin x - \frac{1}{\sqrt{2}} \cos x \right) = -1$$

$$\Rightarrow \sin x \cos 45^\circ - \cos x \sin 45^\circ = \frac{-1}{\sqrt{2}}$$

$$\Rightarrow \sin \left(x - \frac{\pi}{4} \right) = \sin \left(-\frac{\pi}{4} \right)$$

$$= \sin \left(\pi + \frac{\pi}{4} \right) \text{ or } \sin \left(2\pi - \frac{\pi}{4} \right)$$

$$x - \frac{\pi}{4} = -\frac{\pi}{4} \Rightarrow x = 0$$

$$x - \frac{\pi}{4} = \pi + \frac{\pi}{4} \Rightarrow x = \frac{3\pi}{2}$$

$$\text{And } x - \frac{\pi}{4} = 2\pi - \frac{\pi}{4} \Rightarrow x = 2\pi$$

15. (b) As per the given condition.

T_n triangle can be formed by using the vertices of a regular polygon of n .

$$\therefore {}^{n+1}C_3 - {}^nC_3 = 21$$

$$\Rightarrow \frac{(n+1)(n)(n-1)}{1 \cdot 2 \cdot 3} - \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3} = 21 \Rightarrow \frac{n(n-1)}{2} = 21$$

Clearly, $n = 7$

16. (d)

$\bar{x}_1$ and $\bar{x}_2$ are means of two distribution

$$\bar{x}_1 < \bar{x}_2$$

$\bar{x}$ is mean of combined distribution n_1, n_2 number of observation.

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

$$\bar{x} - \bar{x}_1 \Rightarrow \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

$$\bar{x} - \bar{x}_1 \Rightarrow \frac{n_2(\bar{x}_2 - \bar{x}_1)}{n_1 + n_2}$$

$$\bar{x} - \bar{x}_1 > 0 \quad (\because \bar{x}_2 > \bar{x}_1)$$

$$\bar{x} - \bar{x}_1 \dots\dots\dots (1)$$

$$\bar{x} - \bar{x}_2 = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} - \bar{x}_2$$

$$= \frac{n_1(\bar{x}_1 - \bar{x}_2)}{n_1 + n_2}$$

$$\bar{x} - \bar{x}_2 < 0$$

$$\bar{x} < \bar{x}_2 < 0$$

$$\bar{x} < \bar{x}_2 \dots\dots\dots (2)$$

From equation (1) and (2)

$$[\bar{x}_1 < \bar{x} < \bar{x}_2]$$

17. In $\triangle ABC$

BC = 2 (Base)

AO = 1 (height)

Area of $\triangle ABC = \frac{1}{2} \times BC \times AO$

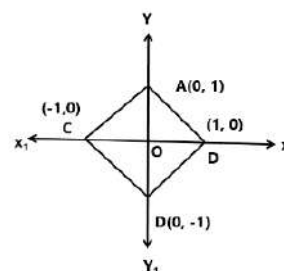
$$= \frac{1}{2} \times 2 \times 1 \text{ sq unit}$$

$\triangle DBC$

$$\text{Area } \triangle DBC = \frac{1}{2} \times 2 \times 1 = 1 \text{ sq unit}$$

$$\text{Area } \square ABCD = 1 + 1$$

$$= 2 \text{ sq. unit}$$



18. (d) $f(x) = lx^2 + mx + n$

As

$$f(1) = f(-1)$$

$$l + m + n = l - m + n$$

Hence, $m = 0$

$$\text{Now, } f(x) = lx^2 + n$$

$$f'(x) = 2lx \Rightarrow f'(a) = 2la$$

$$f'(b) = 2lb, f'(c) = 2lc$$

As a, b, c are in AP, hence $f'(a), f'(b), f'(c)$ are in AP.

19. (b)

$$= \frac{4}{2\sqrt{4x-3}} = \frac{2}{3}$$

$$y = \sqrt{4x-3} - 1 = \frac{dy}{dx} = \frac{4}{2\sqrt{4x-3}}$$

$$\sqrt{4x-3} = 3$$

on squaring both sides, we get

$$4x - 3 = 9$$

$$4x = 12 \Rightarrow x = 3$$

$$\text{and } y = 2$$

Required point is (3, 2)

20. (c)

Probability of Atal speak truth is $\frac{70}{100} = \frac{7}{10}$

Probability of George speak truth is $\frac{40}{100} = \frac{4}{10}$

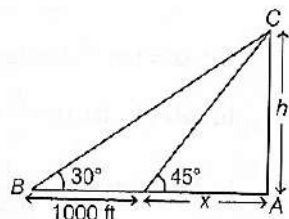
$$\text{Required probability} = \frac{7}{10} \times \frac{4}{10} + \frac{3}{10} \times \frac{6}{10}$$

$$= \frac{28}{100} + \frac{18}{100} = \frac{28+18}{100} = \frac{46}{100} = \frac{23}{50}$$

21. (a) $\tan 45^\circ = \frac{h}{x} \Rightarrow h = x$



and $\tan 30^\circ = \frac{1}{\sqrt{3}} = \frac{h}{x+1000}$
 $\Rightarrow x + 1000 = \sqrt{3}h$



$\Rightarrow x + 1000 = \sqrt{3}x \quad (h = x) \Rightarrow \frac{1000}{\sqrt{3}-1} = x$

$\Rightarrow AB = \frac{1000}{\sqrt{3}-1} + 1000 = 1000 \left(\frac{1+\sqrt{3}-1}{\sqrt{3}-1} \right)$

$= \frac{(\sqrt{3})1000}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = \frac{(\sqrt{3}+1)1000\sqrt{3}}{2}$

$= 500\sqrt{3}(\sqrt{3}+1)\text{ft}$

22. (d) $S_n = 216$ (given)

$S_n = \frac{n}{2} [2a + (n-1)d]$

$\Rightarrow 216 = \frac{n}{2} [2a + (n-1)d]$

$T_n = 2n, a = n$

$2n = n + (n-1)d$

$\Rightarrow d = \frac{n}{(n-1)}$

Then, $216 = \frac{n}{2} \left[2n + (n-1) \times \frac{n}{n-1} \right]$

$\frac{n}{2} [3n] = 216 \Rightarrow \frac{3n^2}{2} = 216$

$\Rightarrow n^2 = \frac{216 \times 2}{3} \Rightarrow n^2 = 72 \times 2 \Rightarrow n^2 = 144 \Rightarrow n = 12$

Putting in Eq. (i)

$d = \frac{12}{11}$

23. (c) Resultant force = $3\hat{i} + 2\hat{j} + 5\hat{k} + 2\hat{i} + \hat{j} - 3\hat{k}$

$F = 5\hat{i} + 3\hat{j} + 2\hat{k}$

$r = 4\hat{i} - 2\hat{j} + 7\hat{k} - 2\hat{i} + \hat{j} + 3\hat{k} \Rightarrow r = 2\hat{i} - \hat{j} + 10\hat{k}$

$W = F \cdot r$

$W = (5\hat{i} + 3\hat{j} + 2\hat{k}) \cdot (2\hat{i} - \hat{j} + 10\hat{k})$

$W = 10 - 3 + 20 = 27 \text{ units}$

24. (a) $9^{\frac{1}{3}} \cdot 9^{\frac{1}{9}} \cdot 9^{\frac{1}{27}} \dots = 9^{\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} \dots} = 9^{1 - \frac{1}{3}} = 9^{\frac{2}{3}} = 3$

25. (c) The minimum value of the function

$f(x) = 2x^3 - 21x^2 + 36x - 20$ is

$f'(x) = 6x^2 - 42x + 36 = 0$

$(x-6)(x-1) = 0$

$x = 6, 1$ are critical point

$f''(x) = 12x - 42$

$f''(6) = 12 \times 6 - 42 = 30 > 0 \rightarrow \text{Minima}$

$f''(1) = 12 \times 1 - 42 = -30 < 0 \rightarrow \text{Maxima}$

There is minima at $x = 6$

$f_{\min} = f(6) = 2 \times 6^3 - 21 \times 36 + 36 \times 6 - 20 = -128$

26. In word 'Corporation' vowels 'OOAIO' consider 1 letter and CRPRTN as 6 letters but R comes 2 times.

Number of ways = $\frac{7!}{2!} = 2520$

In 5 vowels

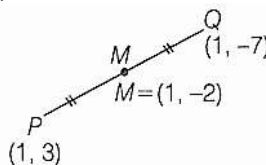
$\downarrow$ O occurs 3 times = $\frac{5!}{3!} = 20$ ways

Required number of ways = $2520 \times 20 = 50400$ ways.

27. (c) $\log_2 x = 10 \Rightarrow x = 2^{10}$

$\log_x y = 100 \Rightarrow y = x^{100} \Rightarrow y = (2^{10})^{100} \Rightarrow y = 2^{1000}$

28. (c) Let equation of line, $2x - 3y = \lambda$ passing through M (mid point of PQ)



$2(1) - 3(-2) = \lambda \Rightarrow \lambda = 8$

$\Rightarrow 2x - 3y = 8 \Rightarrow 2x - 3y - 8 = 0$

29. (c) $\because (a+b+c)(a+b-c) = (a+b)^2 - c^2 = ab$

$\Rightarrow a^2 + b^2 - c^2 + ab = 0 \Rightarrow a^2 + b^2 - c^2 = -ab$

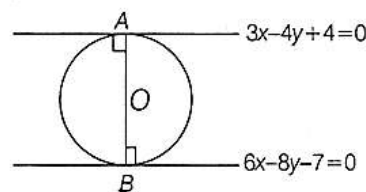
$\cos C = \frac{2(a^2+b^2-c^2)}{2ab} = -1$

$\frac{a^2+b^2-c^2}{2ab} = -\frac{1}{2} \Rightarrow \cos C = -\frac{1}{2} \Rightarrow C = \frac{2\pi}{3}$

30. (d) Total numbers which contains digit 1 = total number of numbers - total numbers which does not contains digit 1 = $10^3 - 9^3$

So, the numbers containing 1 is $10^3 - 9^3$.

31. (b) AB is diameter



$L_1 = 3x - 4y + 4 = 0$

$L_2 = 3x - 4y - 7/2 = 0$

$AB = \frac{\left| \frac{4 - (-7/2)}{3} \right|}{\left| \frac{3}{\sqrt{9+16}} \right|} = \frac{\left| \frac{15}{2} \right|}{\left| \frac{3}{5} \right|} = \frac{3}{2} \left[\because d = \frac{C_1 - C_2}{\sqrt{a^2 + b^2}} \right]$

Hence, radius is $3/4$.

32. (a) If the sides of parallelogram are p and q. Then, their area is $|p \times q|$ = Magnitude of

$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & -2 \\ 1 & -3 & 4 \end{vmatrix}$

= Magnitude of $-2\hat{i} - 14\hat{j} - 10\hat{k}$

$= \sqrt{300} = 10\sqrt{3}$

33. (d)

$\sin x + a \cos x = b$

Let $|a \sin x - \cos x| = k$ (say)

Squaring and adding Eqs. (i) and (ii)

$\Rightarrow \sin^2 x + a^2 \cos^2 x + 2a \sin x \cos x$

$+ \cos^2 x + a^2 \sin^2 x - 2a \sin x \cos x = b^2 + k^2$

$\Rightarrow 1 + a^2 = b^2 + k^2 \Rightarrow k^2 = 1 + a^2 - b^2$

$\Rightarrow k = \pm \sqrt{a^2 - b^2 + 1}$

$\therefore |a \sin x - \cos x| = k = \sqrt{a^2 - b^2 + 1}$

$\tan A - \cot A = \frac{\sin^2 A - \cos^2 A}{\cos A \sin A} = \frac{-2 \cos 2A}{2 \cos A \sin A}$

$\frac{-2 \cos 2A}{\sin 2A} = -2 \cot 2A$

34. (b) $P(A \cup B) = \frac{5}{6}, P(A \cap B) = \frac{1}{3}, P(B) = \frac{1}{2}$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Hence, $P(A) = \frac{4}{6} = \frac{2}{3}$



For independent events

$$P(A \cap B) = P(A) \times P(B)$$

$$1/3 = 2/3 \times 1/2$$

Hence, A and B are independent events.

35. (d) Since, vectors are coplanar

$$\therefore \begin{vmatrix} 2 & -1 & 1 \\ 1 & 2 & -3 \\ 3 & \lambda & 5 \end{vmatrix} = 0$$

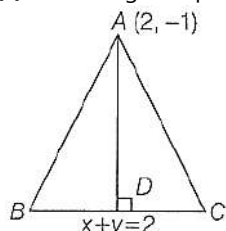
$$\Rightarrow 3(3-2) - \lambda(-6-1) + 5(4+1) = 0 \quad [\text{Expand along } R_3]$$

$$\Rightarrow 3 + 7\lambda + 25 = 0$$

$$\Rightarrow 7\lambda = -28$$

$$\Rightarrow \lambda = -4$$

36. (c) AD is length of perpendicular from vertex A to base BC.



$$\text{In } \triangle ABD, \tan 60^\circ = \frac{AD}{a/2} \Rightarrow AD = \frac{\sqrt{3}a}{2}$$

Where a is side of equilateral triangle

$$AD = \left| \frac{2-1-2}{\sqrt{1+1}} \right| = \frac{1}{\sqrt{2}} = \frac{\sqrt{3}}{2}a \Rightarrow a = \sqrt{\frac{2}{3}}$$

37. (c) Single digit number = 3

Two digit numbers are $3 \times 2 = 6$

Three digit numbers are $3 \times 2 \times 1 = 6$

Hence, total numbers are 15.

38. (a)

$$\frac{dy}{dx} = \cos t \quad \frac{dx}{dt} = \sin t$$

$$\frac{dy}{dx} = \left[\frac{dy/dt}{dx/dt} \right] = \frac{-b}{a} \cot t$$

$$\Rightarrow U = \frac{dy}{dx} = \cot t$$

$$\frac{d^2y}{dx^2} = \frac{du}{dx} = \frac{du}{dt} \times \frac{dt}{dx} > \left(\frac{b}{a} \operatorname{cosec}^2 t \right) \times \left(\frac{-1}{a \sin t} \right)$$

$$= \frac{-b}{a^2 \sin^3 t} = \frac{-b^2}{a^2 y^3}$$

39. (b) Variance = $\Sigma(x_i - \bar{x})^2 P_i$

$$\text{Here, } \bar{x} = \Sigma x_i P_i = 1 \times 0.3 + 2 \times 0.4 + 3 \times 0.3 = 2$$

Variance

$$= (1-2)^2 \times 0.3 + (2-2)^2 \times 0.4 + (3-2)^2 \times 0.3 = 0.6$$

40. (a)

$$\tan A - \cot A = \frac{\sin^2 A - \cos^2 A}{\cos A \sin A} = \frac{-2\cos 2A}{2\cos A \sin A}$$

$$= \frac{-2\cos 2A}{\sin 2A} = -2\cot 2A$$

Add and subtract $\cot \theta$ in given equation.

$$\begin{aligned} \tan \theta - \cot \theta + 2\tan 2\theta + 4\tan 4\theta + 8\cot 8\theta + \cot \theta \\ = -2\cot 2\theta + 2\tan 2\theta + 4\tan 4\theta + 8\cot 8\theta + \cot \theta \\ = 2(\tan 2\theta - \cot 2\theta) + 4\tan 4\theta + 8\cot 8\theta + \cot \theta \\ = 2(-2\cot 4\theta) + 4\tan 4\theta + 8\cot 8\theta + \cot \theta \\ = 4(\tan 4\theta - \cot 4\theta) + 8\cot 8\theta + \cot \theta \\ = 4(-2\cot 8\theta) + 8\cot 8\theta + \cot \theta = \cot \theta \end{aligned}$$

Short Trick:

Let $\theta = 30^\circ$

$$\Rightarrow \tan 30^\circ + 2\tan 60^\circ + 4\tan 120^\circ + 8\cot 240^\circ$$

$$\Rightarrow \tan 30^\circ + 2\tan 60^\circ - 4\tan 60^\circ + 8\tan 30^\circ$$

$$\Rightarrow \frac{1}{\sqrt{3}} + 2\sqrt{3} - 4\sqrt{3} + \frac{8}{\sqrt{3}}$$

$$\Rightarrow \frac{9}{\sqrt{3}} - 2\sqrt{3} = \frac{9-2(3)}{\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3} = \cot 30^\circ$$

41. The numbers between 200 and 400, divisible by 7

203, 210, 217, 399

$$A = 203, d = 7, l = 399$$

$$a_n = a + (n-1)d$$

$$30g, 203 + (n-1)7$$

$$7(n-1) = 196$$

$$(n-1) = 28$$

$$N = 29$$

$$S_{2q} = \frac{n}{2}(a + l)$$

$$= \frac{29}{2}(203 + 399)$$

$$S_{2q} = \frac{29}{2}(602)$$

$$S_{2q} = 29 \times 301$$

$$S_{2q} = 8729$$

The sum of all numbers between 200 and 400 which divided by 7 is 8729.

42. (d) $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^2 \tan x} \left(\frac{0}{0} \text{ form} \right)$

$$\lim_{x \rightarrow 0} \frac{\frac{\sin x}{\cos x} - x}{x^2 \frac{\sin x}{\cos x}} = \lim_{x \rightarrow 0} \frac{\sin x - x \cos x}{x^2 \sin x} \left(\frac{0}{0} \text{ form} \right)$$

Using L' Hospital rule

$$\lim_{x \rightarrow 0} \frac{\cos x + x \sin x - \cos x}{2x \sin x + x^2 \cos x} = \lim_{x \rightarrow 0} \frac{\sin x}{2 \sin x + x \cos x} \left(\frac{0}{0} \text{ form} \right)$$

Again using L' Hospital rule,

$$\lim_{x \rightarrow 0} \frac{\cos x}{2 \cos x - x \sin x + \cos x} = \frac{1}{3}$$

43. (b)

$$\text{Clearly, } n(S) = (6 \times 6) = 36$$

Let E = Event that the sum is a prime number

Then $E = \{(1,1), (1,2), (1,4), (1,6), (2,1), (2,3), (2,5),$

$(3,2), (3,4), (4,1), (4,3), (5,2), (5,6), (6,1), (6,5)\}$ $n(E) = 15$

Hence, the probability that the total score is a prime number is

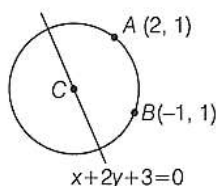
$$\frac{15}{36} = \frac{5}{12}$$

44. (a) Let (α, β) is the centre of circle which lies on the line

$$x + 2y + 3 = 0$$

$$\Rightarrow \alpha + 2\beta + 3 = 0 \Rightarrow \beta = \frac{-\alpha-3}{2}$$

Hence, centre C $\left(\alpha, \frac{-\alpha-3}{2} \right)$



$$CA = CB = \text{radii}$$

$$CA^2 = CB^2$$



$$\Rightarrow (\alpha - 2)^2 + \left(\frac{-\alpha-3}{2} - 1\right)^2 = (\alpha + 1)^2 + \left(\frac{-\alpha-3}{2} - 1\right)^2$$

$$\Rightarrow \alpha^2 + 4 - 4\alpha = \alpha^2 + 1 + 2\alpha$$

$$\Rightarrow 6\alpha = 3$$

$$\Rightarrow \alpha = \frac{1}{2}$$

$$\therefore C\left(\frac{1}{2}, \frac{-7}{4}\right)$$

$$\text{Radius} = \sqrt{\left(2 - \frac{1}{2}\right)^2 + \left(1 + \frac{7}{4}\right)^2} = \frac{\sqrt{157}}{4}$$

$$\text{Equation } \left(x - \frac{1}{2}\right)^2 + \left(y + \frac{7}{4}\right)^2 = \frac{157}{16}$$

$$x^2 + \frac{1}{4} - x + y^2 + \frac{49}{16} + \frac{7}{2}y = \frac{157}{16}$$

$$2x^2 + 2y^2 - 2x + 7y - 13 = 0$$

45. (b) Area = $\frac{1}{2}|a \times b + b \times c + c \times a|$

46. (c) Focal property $PF' + PF = 2a$

47. (b)

$$\sin x + \sin^2 x = 1 \Rightarrow \sin x = \cos^2 x$$

$$\cos^2 x + \cos^4 x = (\sin x) + (\sin x)^2 = 1$$

48. (b)

The probability of success is twice the probability of failure.

Let the probability of failure be p .

Probability of success = $2p$

$$p + 2p = 1$$

$$\Rightarrow 3p = 1$$

$$p = \frac{1}{3}$$

Hence probability of getting success is $\frac{2}{3}$ and failure is $\frac{1}{3}$.

Probability = Probability of getting atleast 4 success.

$$= {}^6C_4 \left(\frac{2}{3}\right)^4 \left(\frac{1}{3}\right)^2 + {}^6C_5 \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right) + {}^6C_6 \left(\frac{2}{3}\right)^6$$

$$= \frac{496}{729}$$

49. (c) $-1^2 + 2^2 - 3^2 + 4^2 + \dots (20)^2$ (given)

$$\Rightarrow \{(2)^2 - (1)^2\} + \{(4)^2 - (3)^2\} + \dots + \{(20)^2 - (19)^2\}$$

$$\Rightarrow 3 + 7 + 11 + \dots + 39$$

Required sum for (10 terms)

$$\frac{n}{2}(a + l) = 10 \times \left(\frac{3+39}{2}\right) = 210$$

50. (b)

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\Rightarrow \tan(\alpha + \beta) = \frac{\frac{m}{1-m} + \frac{1}{2m+1}}{1 - \frac{m}{1-m} \cdot \frac{1}{2m+1}} = \frac{m(2m+1)+m+1}{(m+1)(2m+1)-m}$$

$$= \frac{2m^2+m+m+1}{2m^2+3m+1-m} = 1$$

$$\tan(\alpha + \beta) = 1$$

$$\Rightarrow \alpha + \beta = \tan^{-1}(1) = \frac{\pi}{4}$$

REASONING

51. (c) Let the length of the train be x m.

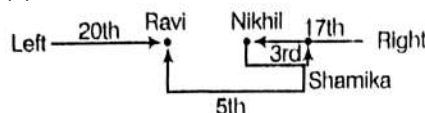
$$\text{Then, } \frac{x+162}{18} = \frac{x+120}{15}$$

$$\Rightarrow \frac{x+162}{6} = \frac{x+120}{5}$$

$$\Rightarrow 6x + 720 = 5x + 810$$

$$\Rightarrow x = 90 \text{ m}$$

52. (b)

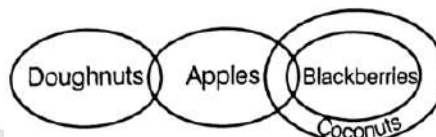


$\therefore$ Ravi's position from the left = 20th

and Ravi's position from the right = $17 + 2 = 19$ th

$\therefore$ Number of children in the row = $(20 + 19) - 1 = 38$

53. (a) According to the statements,



Hence, some blackberries are doughnuts, is definitely, false.

Solutions [Q. Nos. 54 to 56]

Reena's mother-in-law $\rightarrow$ head of the family $\rightarrow$ Lawyer

Rakesh $\rightarrow$ Married $\rightarrow$ Teacher

Mukesh $\rightarrow$ Married $\rightarrow$ Teacher

Reena $\rightarrow$ Rakesh's wife $\rightarrow$ Lawyer

Mukesh's wife $\rightarrow$ Doctor

Ajay $\rightarrow$ Mukesh's son

Ajay's grandfather $\rightarrow$ Unknown profession

54. (a) Rakesh (Teacher) and Reena (Lawyer) is definitely a couple.

55. (c) Rakesh's wife's profession is Lawyer.

56. (d) We cannot determine the Ajay's grandfather occupation.

57. (b) The pattern of series is as follows:

$$A \xrightarrow{(B)} CD \xrightarrow{(EF)} GHI \xrightarrow{(J,K,L)} MNOP \xrightarrow{(Q,R,S,T)} UVWXY$$

58. (b) As,

$$\begin{array}{ll} F \xrightarrow{+1} G & \text{Similarly, } G \xrightarrow{+1} H \\ R \xrightarrow{+2} T & U \xrightarrow{+2} W \\ I \xrightarrow{+3} L & A \xrightarrow{+3} D \\ E \xrightarrow{+4} J & R \xrightarrow{+4} V \\ N \xrightarrow{+5} S & D \xrightarrow{+5} I \\ O \xrightarrow{+6} J & \end{array}$$

59. (d) Given, Alan's age = 34 yr

$$\text{Carl's age} = 34 + 2 = 36 \text{ yr}$$

$$\text{Bob's age} = 36 - 1 = 35 \text{ yr}$$

$$\text{Dave's age} = 35 + 2 = 37 \text{ yr}$$

Hence, Dave is the oldest among them.

Solutions (Q. Nos. 60 to 62)

1	2	3	4	5	6

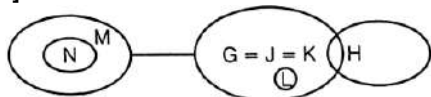
(Mr. Donald) (Mr. Tim) (Mr. Mike) (Mr. Brown) (Miss. Hardy) (Miss. Robert)
Smoker smoker smoker Non-smoker Non-smoker Non-smoker

60. (d) Mr. Donald's office is farthest from Mr. Brown.



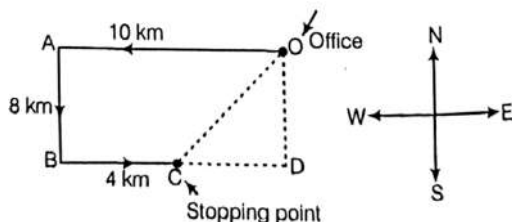
61. (d) The three smoker employers of the offices should be seated in 1, 2 and 3.
62. (d) The ideal office for Mr. Mike would be 3.
63. (a) Doctor's previous time to see the patients
 $= 1:40 \text{ pm} - 3 \text{ h } 30 \text{ m} = 10:10 \text{ am}$
 So, the information given to the compounder by the doctor
 $= 10:10 \text{ am} + 1:20 - 11:30 \text{ am}$
64. (c) Series is increasing in three consecutive terms like
 42 40 38; 35 33 31; 28 26 24

Solutions [Q. Nos. 15 to 16]



65. (b) If no P's are K's, then if any P is a G, it is a J, must be, true.
66. (c) Some H's are both M's and G's is inconsistent with one or more of the conditions.
67. (d) According to the question,
 Shyam > Pradeep = Anurag Suresh > (Anand, Praveen)
 Hence, Shyam is tallest among them.
68. (d) Suppose,
 Rajeev's age = 0 (Born)
 Then, Rajeev's brother's age = 6yr
 Rajeev's father's age = 6 + 32 = 38 yr
 Rajeev's mother's age = 38 - 3 = 35 yr
 Rajeev's sister's age = 35 - 25 = 10 yr

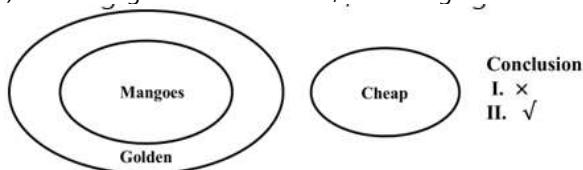
69. (c) Dhoni's walking directions are as follows:



$$\begin{aligned} \therefore OD &= AB = 8 \text{ km} \\ \Rightarrow CD &= BD(AO) - BC \\ &= (10 - 4) \text{ km} \\ &= 6 \text{ km} \\ \therefore \text{Required distance (OC)} &= \sqrt{OD^2 + CD^2} \\ &= \sqrt{8^2 + 6^2} = \sqrt{64 + 36} \\ &= \sqrt{100} = 10 \text{ km} \end{aligned}$$

70. (a) By option, number 58 follows the statements rule.
- 583 × 19 + 1
- 584 × 14 + 2
- 585 × 11 + 3
- 586 × 9 + 4

71. (b) According to the statements, Venn diagram is as follows:



Hence, only conclusion II follows.

Solutions [Q. Nos. 22 to 23]

According to the question.

$$\begin{aligned} A &= 2B && \dots(i) \\ B &= 4.5 C && \dots(ii) \\ C &= \frac{1}{2} D && \dots(iii) \\ D &= \frac{1}{2} E && \dots(iv) \\ A &> E > C && \dots(v) \end{aligned}$$

Combining all the equations,
 $A > B > E > D > C$

72. (a) A is the heaviest article in weight.
73. (a) A, B, E, D, C

Solutions [Q. Nos. 74 to 76]

$$\begin{aligned} A \text{ (only ON)} &B \text{ (ON)} && \dots(i) \\ A \text{ (ON)} + B \text{ (ON)} &\rightarrow C \text{ (ON)} && \dots(ii) \\ A \text{ (ON)} + B \text{ (ON)} + C \text{ (ON)} &\rightarrow C \text{ (OFF)} && \dots(iii) \end{aligned}$$

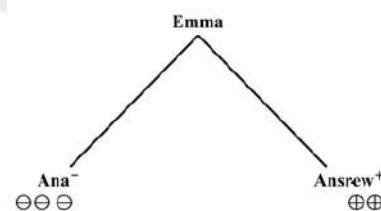
Except,
 $ON \rightarrow OFF$ and $OFF \rightarrow ON$

74. (b) If switches A and B are ON and C is OFF, their changed settings will be A ON, B ON and C ON.
75. (c) If only B is ON, the changed setting will be A ON, B OFF and C ON.
76. (b) If only B is ON in the changed setting, the original setting could have been A ON, B OFF and C ON.
77. (c) $\therefore$ Fourth date after 21st = 21 + 4 = 25
 Then,

3	10	17
↑	↑	↑
1st Friday	2nd Friday	3rd Friday
4th Friday		

$\therefore$ Required day (25th date) = Friday + 1 = Saturday

78. (d)



$$\therefore \text{Number of children} = \ominus + \ominus + \ominus + \text{Ana}^- + \text{Andrew}^+ + \oplus + \oplus = 7$$

Solutions [Q. Nos. 79 to 81]

$$\begin{aligned} \text{Anu} &> \text{Cini} && \dots(i) \\ \text{Binu} &> \text{Eenu} && \dots(ii) \\ \text{Dany} &> \text{Anu} && \dots(iii) \\ \text{Eenu} &> \text{Anu} && \dots(iv) \end{aligned}$$

Combining all the equations,

$$(\text{Dany, Binu}) > \text{Eenu} > \text{Anu} > \text{Cini}$$

79. (c) Dany or Binu is the tallest among them.
80. (a) Cini is the shortest among them.

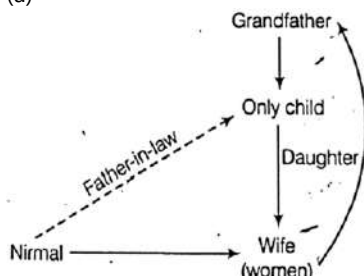


81. (d) Dany is the tallest in the group, which would help to logically order the persons according to their heights.

82. (d) In 1st 100 m race, distance covered by Karan = 100 m
and distance covered by Arjun = 90 m
In 2nd 100 m race, distance covered by Karan = 110 m
and distance covered by Arjun = ?
100 m of Karan = 90 m of Arjun
110 m of Karan = $\frac{90 \times 110}{100} = 99$ m of Arjun
 $\therefore$ Karan beats Arjun by 1 m.

83. (c) India (+2) $\rightarrow$ Australia (-2)
West Indies (+2) $\rightarrow$ India (-2)
[+ $\rightarrow$ win]
[- $\rightarrow$ loss]
Australia (+2) $\rightarrow$ West Indies (-2)
India (+2) New Zealand (-2)
West Indies (+2) $\rightarrow$ New Zealand (-2)
Hence, New Zealand Country has lost 4 matches most number of times.

84. (a)



So, women is related as wife of the Nirmal.

Solutions [Q. Nos. 85 to 87]

Condition I	Condition II	Condition III	Condition IV
6-C	6-	6... B	6... B
5-	5-C	5... D	5. B.D
4-A	4-	4-C	4. D
3... B	3-A	3-	3-C
2... B, D	2. B	2-A	2-
1-D	1... D	1-	1-A

85. (d) According to condition IV, if A is standing on step 1, then D is standing one step higher than C.

86. (c) According to conditions I and II, if D is standing on step, 1 then A could be standing on either step 3 or 4 only.

87. (b) According to conditions I, III and IV, it there are two steps in between the steps A and D, then A may be standing in 4, 2 and 1 steps. So, in respect of option, we say that A must be standing on step 4.

88. (b) $A \times B = A = B$

$$A - B \Rightarrow A > B$$

$$A + B \Rightarrow A < B$$

Then, Sachin * Mohan - Ravi

$$\Rightarrow \text{Sachin} = \text{Mohan} > \text{Ravi}$$

Hence, Ravi is youngest among them.

89. (a) Total movement of hour hand = $20^\circ + 20^\circ = 40^\circ$

So, hour hand covers $\frac{1^\circ}{2}$ in 1 min.

Also, hour hand covers 40° in 80 min.

Time duration to go tennis court and returning in

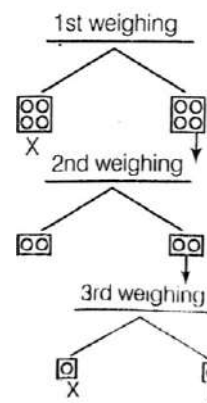
(10+10) min

= 20 min.

$\therefore$ Total spent time at the tennis court = (80 - 20) min

= 60 min

90. (c)



Hence, three times of weighing is required to identify the heavier ball.

ENGLISH

91. d 92. c 93. d 94. a 95. b 96. b 97. c
98. b 99. d 100. d 101. d 102. b 103. a 104. c
105. c 106. a 107. d 108. a 109. c 110. c

COMPUTER

111. (b)

All digital circuits can be realized by using only half adders. A half adder consists of an XOR gate and an AND gate.

112. (b)

The Boolean function $a + (a' b)$ is solved as

$$a + (a' + b)$$

$$= (a' + a') \cdot (a + b) \quad [\text{using distributive law}]$$

$$= 1 \cdot (a + b) \quad [\because a + a' = 1]$$

$$= a + b$$

113. (a)

The flip-flop circuit is used as a memory device in computers.

114. (b)

Conversion of hexadecimal 4DF to its octal as

$$\begin{array}{ccc} 4 & D & F \\ | & | & | \\ 0100 & 1101 & 1111 \end{array} = (010011011111)_{16}$$

Now, $(010011010000)_{16}$ into octal number is

$$\frac{010}{2} \frac{011}{3} \frac{011}{3} \frac{111}{7} = (2337)_8$$

115. (c)

As we known a tautology is a Boolean formula that is always true.

From the giving options we solve all through the truth table.

Here $x + \bar{y} + \bar{x}$ is satisfying the condition of tautology.

x	y	$\bar{x}$	$\bar{y}$	$x + \bar{y} + \bar{x}$
0	0	1	1	1
0	1	1	0	1
1	0	0	1	1
1	1	0	0	1

The output of $x + \bar{y} + \bar{x}$ is true.



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116. (c)

EEPROM stands for Electrically Erasable programmable Read only Memory. It is a type of a non-volatile memory used in computers and other electronic devices.

117. (d)

Compact Disc (CD) is a disc which contains digital data that is scanned by a laser beam for the reproduction of recorded sound.

118. (b)

WWW stands for World Wide Web.

119.(c)

The multiplication of $(10111)_2 \times (1110)_2$ is

$$(10111)_2 \times (1110)_2 = (000101000010)_2$$

Now, the result is converted into hexadecimal as

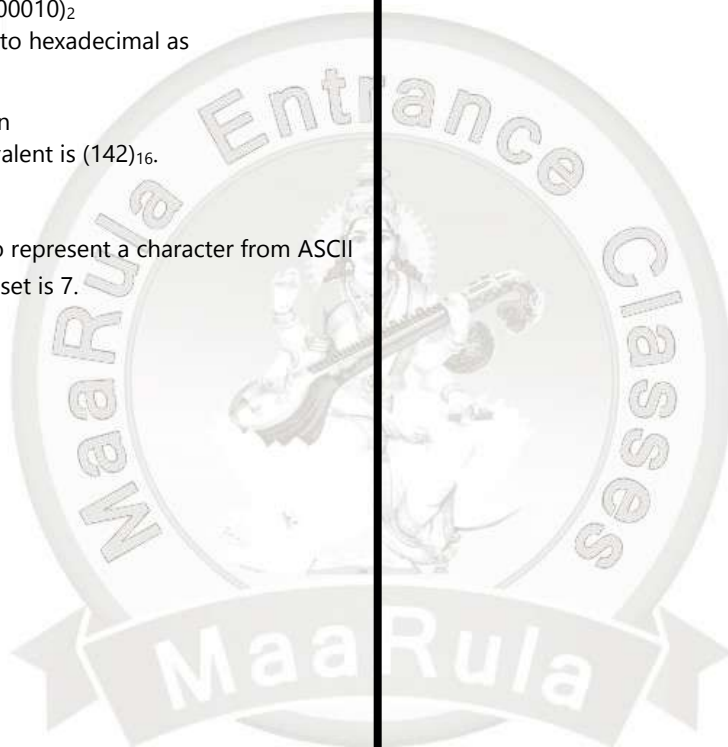
$$\begin{array}{ccc} 0001 & 0100 & 0010 \\ \hline 1 & 4 & 2 \end{array}$$

So, the answer of the operation

$(10111)_2 \times (1110)_2$ in hex equivalent is $(142)_{16}$.

120.(d)

The minimum number of bits to represent a character from ASCII code set is 7.



ADDRESS: 117/466, O Block, Geeta Nagar, Sharda Nagar, NEAR: ANURAG HOSPITAL Kanpur, Uttar Pradesh 208025



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